Chapter 2: Vectors and Tensors

Reading: [Fung/94] Chapter 1

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Contents

(§2.1) Vector Algebra

Objectives

- To review the basic operations in vector algebra

Definition

- A directed line segment with a given magnitude and direction
- In reference with a coordinate system of measurement, a vector is an ordered list of triple (three-dimensional) or double (two-dimensional) numbers

$$\{1, -1, 3\} \qquad \{\cos\theta, \sin\theta\}$$

Notation

- In printed form, either the overline, $\vec{AB}$, or lowercase **boldface** *italic* font will be used to denote vectors.

$$\boldsymbol{u} = \vec{AB}, \quad \boldsymbol{e}_2, \quad \boldsymbol{f}$$

- A zero vector $\mathbf{0}$ has zero magnitude.
- Vector magnitudes are indicated by the symbols $|\boldsymbol{u}|$, or simply as u, using the same letter without **boldface** *italic*.

- Multiplying a vector by a scalar scales the vector magnitude but has no effect on it's magnitude, e.g., multiplying $\boldsymbol{u}$ with the reciprocal of its magnitude, i.e.

$$\left(\frac{1}{|\boldsymbol{u}|}\right)\boldsymbol{u} = \frac{\boldsymbol{u}}{|\boldsymbol{u}|} = \frac{\boldsymbol{u}}{u} = \boldsymbol{e}_u$$

  gives a vector of length unity. Note that, $\boldsymbol{e}_u$ is a *base vector.*

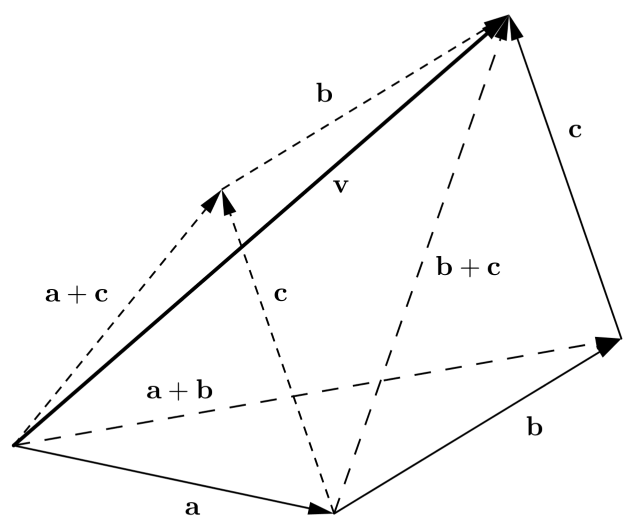
- A unit vector is a vector of magnitude unity (1). In this course (and in many others), the letter $\boldsymbol{e}$ is reserved to denote unit vectors, e.g.

$$\boldsymbol{e}_1,\ \boldsymbol{e}_u,\ \boldsymbol{e}_\theta, \ldots$$

- We can add two or more vectors using the + operator

$$\boldsymbol{a} + \boldsymbol{b}$$

  which is known as the parallelogram rule.



  - Subtraction can be formulated by combining negation and summation

$$\boldsymbol{a} - \boldsymbol{b} = \boldsymbol{a} + (-\boldsymbol{b})$$

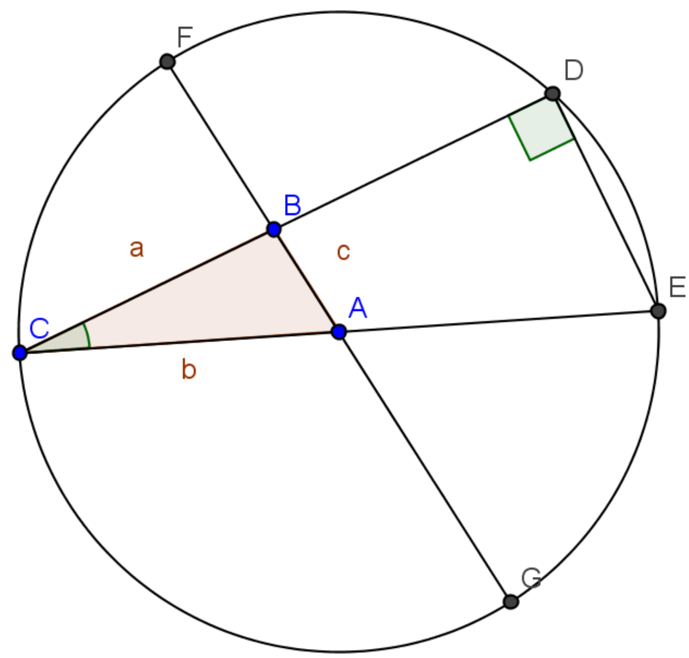
The Law of Sines and Cosines

- The vectors: $\boldsymbol{a} = \vec{BC}$, $\quad \boldsymbol{b} = \vec{CA}$, $\quad \boldsymbol{c} = \vec{BA} = \boldsymbol{a} + \boldsymbol{b}$
- the corner angles $C = (\boldsymbol{a}, \boldsymbol{b})$, $A = (\boldsymbol{b}, \boldsymbol{c})$, $B = (\boldsymbol{a}, \boldsymbol{c})$
- The **Law of Cosines** reads

$$c^2 = a^2 + b^2 - 2ab \cos C$$

- The **Law of Sines** reads

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



Scalar (dot) Product

- The *scalar* (or *dot*) *product* of $\boldsymbol{u}$ and $\boldsymbol{v}$, denoted $\boldsymbol{u} \cdot \boldsymbol{v}$, is defined by the formula

$$\boldsymbol{u} \cdot \boldsymbol{v} = |\boldsymbol{u}||\boldsymbol{v}| \cos \theta \quad (0 \le \theta \le \pi),$$

where $\theta = (\boldsymbol{u}, \boldsymbol{v})$ is the angle between the given vectors.

- One of the major uses of this operation is to determine the projected length, $u_{\|a}$, of a vector, $\boldsymbol{u}$, along an axis, $a - a$ in space.
  - A unit vector along the axis $a - a$ is constructed by locating two points C and D on $a - a$, constructing the vector $\vec{CD}$, and then dividing it to its length $|\vec{CD}|$

$$\boldsymbol{e}_a = \vec{CD}/|\vec{CD}|$$

  - The dot product between $\boldsymbol{u}$ and the unit vector yields the projection, also called *component* $\boldsymbol{u}_{\|a}$

$$\boldsymbol{u} \cdot \boldsymbol{e}_a = u_{\|a}$$

- Introducing a coordinate system $O - x_1x_2x_3$, using the indices {1,2,3} to denote the first, second and third coordinate directions having a set of unit **base vectors** $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, and $\boldsymbol{e}_3$, respectively.

- We can obtain a representation of the vector, $\boldsymbol{u}$, in terms of its components by using the *dot product* operation

$$u_1 = \boldsymbol{u} \cdot \boldsymbol{e}_1, \quad u_2 = \boldsymbol{u} \cdot \boldsymbol{e}_2, \quad u_3 = \boldsymbol{u} \cdot \boldsymbol{e}_3,$$

  which can be written in shorthand (indicial) notation by the use of a *free-index* i

$$u_i = \boldsymbol{u} \cdot \boldsymbol{e}_i, \quad i = 1, 2, 3$$

- In this way, the representation of the vector in the $O - x_1x_2x_3$ system can be written as

$$\boldsymbol{u} = u_1\boldsymbol{e}_1 + u_2\boldsymbol{e}_2 + u_3\boldsymbol{e}_3 = \sum_{i=1}^{3} u_i\boldsymbol{e}_i$$

  where summation occurs over a *repeated index* i.

- Note from it's definition the *dot product* of two perpendicular vectors $(\boldsymbol{u}, \boldsymbol{v}) = \pi/2$ yields

$$\boldsymbol{u} \cdot \boldsymbol{v} = |\boldsymbol{u}||\boldsymbol{v}| \cos(\pi/2) = 0$$

- This can be used to obtain the well-known relation between the base vectors

$$\boldsymbol{e}_1 \cdot \boldsymbol{e}_1 = 1, \quad \boldsymbol{e}_1 \cdot \boldsymbol{e}_2 = 0, \quad \boldsymbol{e}_1 \cdot \boldsymbol{e}_3 = 0 \tag{1}$$

$$\boldsymbol{e}_2 \cdot \boldsymbol{e}_1 = 0, \quad \boldsymbol{e}_2 \cdot \boldsymbol{e}_2 = 1, \quad \ldots, \qquad \text{etc.} \tag{2}$$

- A shorthand of the nine equations above is provided by introducing the *free-indices* i, j as

$$\boldsymbol{e}_i \cdot \boldsymbol{e}_j = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases} \tag{3}$$

  where i, j can freely take any values from the set of coordinate numbers $1, 2, 3$.

- This above gives rise to the definition of the *Kroenecker's delta*, δ_{ij}

$$\delta_{ij} = \boldsymbol{e}_i \cdot \boldsymbol{e}_j$$

  which is an *identity matrix* in a sense.

Review Problems

2.1 Given vector $\mathbf{u} = -3\mathbf{e}_1 + 4\mathbf{e}_2 + 5\mathbf{e}_3$, find a unit vector in the direction of $\mathbf{u}$.

Answer: $(\sqrt{2}/10)\mathbf{u}$.

2.2 If $\overrightarrow{AB} = -2\mathbf{e}_1 + 3\mathbf{e}_2$, and the midpoint of the segment $\overrightarrow{AB}$ has coordinates $(-4, 2)$, find the coordinates of A and B.

Answer: $(-3, \frac{1}{2}), (-5, \frac{7}{2})$.

2.3 Prove that, for any two vectors $\mathbf{u}, \mathbf{v}$, $|\mathbf{u} - \mathbf{v}|^2 + |\mathbf{u} + \mathbf{v}|^2 = 2(|\mathbf{u}|^2 + |\mathbf{v}|^2)$.

2.4 Find the magnitude and direction of the resultant force of three coplanar forces of 10 lb each acting outward on a body at the origin and making angles of 60°, 120°, and 270°, respectively, with the x-axis.

Answer: $10(\sqrt{3} - 1), \perp x$.

2.5 Find the angles between $\mathbf{u} = 6\mathbf{e}_1 + 2\mathbf{e}_2 - 3\mathbf{e}_3$ and $\mathbf{v} = -\mathbf{e}_1 + 8\mathbf{e}_2 + 4\mathbf{e}_3$.

Answer: $\cos^{-1}(-\frac{2}{63})$.

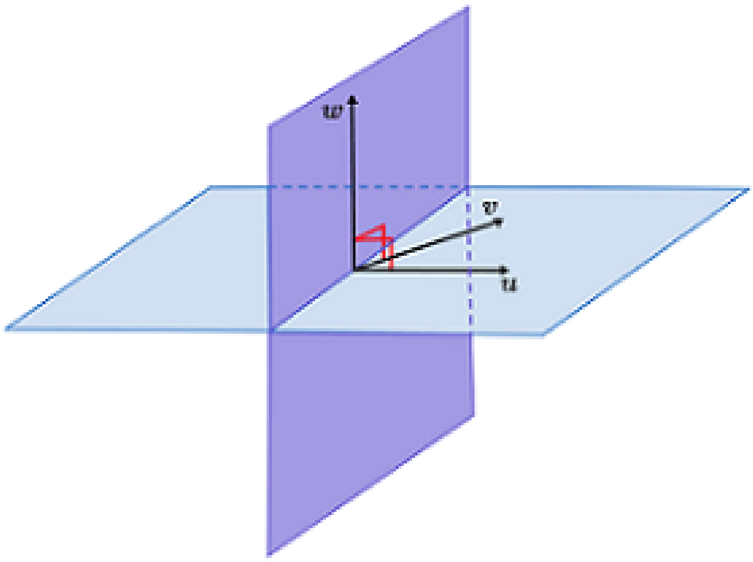
2.6 Given $\mathbf{u} = 3\mathbf{e}_1 + 4\mathbf{e}_2 - \mathbf{e}_3$, $\mathbf{v} = 2\mathbf{e}_1 + 5\mathbf{e}_3$, find the value of α so that $\mathbf{u} + \alpha\mathbf{v}$ is orthogonal to $\mathbf{v}$.

Vector (cross) Product

- Whereas the scalar product of two vectors is a scalar quantity, the *vector* (or *cross*) *product* of two vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ produces another vector $\boldsymbol{w}$; and we write $\boldsymbol{w} = \boldsymbol{u} \times \boldsymbol{v}$. The magnitude of $\boldsymbol{w}$ is defined as

$$|\boldsymbol{w}| = |\boldsymbol{u}||\boldsymbol{v}| \sin\theta \quad (0 \leq \theta \leq \pi)$$

where $\theta = (\boldsymbol{u}, \boldsymbol{v})$, and the vector $\boldsymbol{w}$ is directed perpendicular to the plane formed by $\boldsymbol{u}$ and $\boldsymbol{v}$, in such a way that $\boldsymbol{u}$, $\boldsymbol{v}$, $\boldsymbol{w}$ form a right-handed screw.



- Vector products satisfy the following relations:

$$\boldsymbol{u} \times \boldsymbol{v} = -(\boldsymbol{v} \times \boldsymbol{u}) \tag{4}$$

$$\boldsymbol{u} \times (\boldsymbol{v} + \boldsymbol{w}) = \boldsymbol{u} \times \boldsymbol{v} + \boldsymbol{u} \times \boldsymbol{w} \tag{5}$$

$$\boldsymbol{u} \times \boldsymbol{u} = \boldsymbol{0} \tag{6}$$

- Very useful in finding the moment of physical variables, such as *moment of a force*, or the *moment of momentum about a point.*

- Calculations become possible upon designation of a set of inertial or relative coordinates.

The vector product can be expressed in terms of the vector components in a coordinate system $O - x_1x_2x_3$ as

$$\boldsymbol{u} \times \boldsymbol{v} = (u_1\boldsymbol{e}_1 + u_2\boldsymbol{e}_2 + u_3\boldsymbol{e}_3) \times (v_1\boldsymbol{e}_1 + v_2\boldsymbol{e}_2 + v_3\boldsymbol{e}_3)$$

- This can be expanded into nine terms (on the board) ...

- which finally yield

$$\boldsymbol{u} \times \boldsymbol{v} = (u_2 v_3 - u_3 v_2)\boldsymbol{e}_1 + (u_3 v_1 - u_1 v_3)\boldsymbol{e}_2 + (u_1 v_2 - u_2 v_1)\boldsymbol{e}_3$$

- The expression can be written in a compacted for by using the concept of a *determinant*

$$\boldsymbol{u} \times \boldsymbol{v} = \begin{vmatrix} cu_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ \boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3 \end{vmatrix}$$

It may be instructive to revisit the ideas of *combinations* and *permutations*

- Game theory?

- Ask GPT:

  > can you summarize the notions of combination and permutation in the context of set logic? can you demonstrate this definition by applying it to the notion of a determinant of a matrix, for example?

- It is interesting to discover certain symmetries in the determinant that govern how the triplets u_i, v_j and $\boldsymbol{e}_k$ permute and combine with each other.

- For example, looking at the first paranthesis, the term $u_2 v_3 \boldsymbol{e}_1$ has an index triplet 231 with a + sign while the next term $u_3 v_2 \boldsymbol{e}_1$ with an index triplet 321 has a − sign.

- We notice that positive $\circlearrowleft$ cyclic permutations (123, 231, 312) yield positive coefficients, while negative $\circlearrowright$ cyclic permutations (321, 213, 132) yield the opposite sign (negative).

- Also, none of the triplets that appear in the result contain a repeating index (e.g. 221,333, etc.)

- In this way, a novel apparatus called the *alternating symbol* was invented (by Levi-Civita) to give the sign of permuting triplets

$$\boldsymbol{\epsilon}_{ijk} = \begin{cases} +1, & \circlearrowleft \text{ cycles} \quad (123, 231, 312) \\ -1, & \circlearrowright \text{ cycles} \quad (321, 213, 132) \\ \quad 0 & \text{otherwise(repeating)} \end{cases} \tag{7}$$

- The result of the cross product operation can be written in shorthand (indicial) notation by multiplying the free-triple-index term $u_i v_j \boldsymbol{e}_k$ with the alternating symbol and summing over all the free indices

$$\boldsymbol{u} \times \boldsymbol{v} = \sum_k \sum_j \sum_i \epsilon_{ijk} u_i v_j \boldsymbol{e}_k$$

- The resulting sum has 27 terms, only six of which are non-zero. Three of these (the $\circlearrowright$ cycles) appear with a $+$ sign while the other three (the $\circlearrowleft$ cycles) with a $-$ sign.

- Note that, similar to the expression in the dot product, summations take place over three pairs of *repeated indices* again.

Review Problems

2.7 Given $\mathbf{u} = 2\mathbf{e}_1 + 3\mathbf{e}_2$, $\mathbf{v} = \mathbf{e}_1 - \mathbf{e}_2 + 2\mathbf{e}_3$, $\mathbf{w} = \mathbf{e}_1 - 2\mathbf{e}_3$, evaluate $\mathbf{u}\cdot(\mathbf{v} \times \mathbf{w})$ and $(\mathbf{u} \times \mathbf{v})\cdot\mathbf{w}$.

Answer: 16.

2.8 $(\mathbf{u} \times \mathbf{v})\cdot\mathbf{w}$ is called the *scalar triple product* of $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$. Show that $(\mathbf{u} \times \mathbf{v})\cdot\mathbf{w} = \mathbf{u}\cdot(\mathbf{v} \times \mathbf{w})$.

2.9 Find the equation of the plane through $A(1, 0, 2)$, $B(0, 1, -1)$, and $C(2, 2, 3)$.

Answer: $7x - 2y - 3z - 1 = 0$.

2.10 Find the area of ΔABC in Prob. 2.9.

Answer: $\sqrt{62}/2$.

2.11 Find a vector perpendicular to both $\mathbf{u} = 2\mathbf{e}_1 + 3\mathbf{e}_2 - \mathbf{e}_3$ and $\mathbf{v} = \mathbf{e}_1 - 2\mathbf{e}_2 + 3\mathbf{k}$.

Answer: $7\mathbf{e}_1 - 7\mathbf{e}_2 - 7\mathbf{e}_3$.

(§2.2) Vector Calculus

Vector Equations

- An important difference between an equation in terms of *scalars* and an equation in terms of *vectors* is the amount of information within.

- For example, in order to establish the identity of two vectors

$$\boldsymbol{M} = \boldsymbol{r} \times \boldsymbol{F}$$

has to be satisfied with respect to each and every one of the components of the resulting vectors on each side (i.e. two for a 2-D space, of three for a 3-D space)

- Consider the conservation of momentum

$$\frac{d}{dt} m\boldsymbol{v} = \sum_i \boldsymbol{F}^i$$

essentially provides three equations in a 3-D space

Or consider describing a plane with normal $\boldsymbol{n}$ that is at a distance p from a point O.

- We can show that the position vector $\boldsymbol{r}$ for any point P that starts from O and end up in this plane satisfies $\boldsymbol{r} \cdot \boldsymbol{n} = p$.

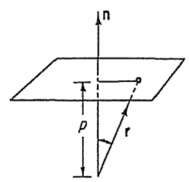


Figure 2.1 Equation of a plane, $\mathbf{r}\cdot\mathbf{n} = p$.

- The vector equation above can be expressed in terms of the vector components with respect to a fixed frame of reference $O - x_1x_2x_3$.

$$n_1x_1 + n_2x_2 + n_3x_3 - p = 0$$

- Yet this eqaution can further be simplified by relying on the notion of a *free-index*

$$\sum_i n_ix_i - p = 0$$

Definition of a Vector Field

- Say a physical variable $\boldsymbol{v}$ measured at points of space exhibits changes as a function of the point coordinates, $\boldsymbol{x}$, as well as time, t.

- Consider the atmospheric current visualization provided by an app (such as `https://windy.com`)

Derivatives of a Vector Field

- A vector field of the air current velocities is defined by

$$\boldsymbol{v} = \boldsymbol{v}(\boldsymbol{x}, t)$$

- This implies the existence of derivatives with respect to the vector arguments (position and time)
- For example, the time derivative of the vector field would be like

$$\frac{\partial \boldsymbol{v}}{\partial t}$$

- The derivative with respect to spatial coordinates, x_i, would be like

$$\frac{\partial \boldsymbol{v}}{\partial x_i}$$

  called the *spatial derivative.* Here i represents the *free-index.*

Curvilinear Coordinate Systems

- Apart from **linear** rectangular coordinates (such as Cartesian), other ways of measuring location of points exist, such as, the spherical, *cylindrical*, parabolic, elliptic, etc. systems. These are known, in general, as **curvilinear** coordinates.
- The **cylindrical** coordinate system, for example, is defined by an axis of revolution, usually labeled with z, an origin O, and a fixed plane through the z axes used in measuring the angular orientations of points in space.
- The measurements in the cylindrical coordinates are labeled as $\{r,\, \theta,\, z\}$ which correspond with bases vectors $\{\boldsymbol{e}_r,\, \boldsymbol{e}_\theta,\, \boldsymbol{e}_z\}$.
- The **polar** coordinate system is a reduced version of the cylindrical system, obtained by eliminating the axial (z) coordinate. Hence, the coordinates and their base vectors in the polar system are $\{r,\, \theta\}$ and $\{\boldsymbol{e}_r,\, \boldsymbol{e}_\theta\}$, respectively.

Another Application of the Cross Product

- The orbital plane motion of a particle, P, located at a fixed distance r away from an axis, $z - z$, about which P rotates with an angular velocity, ω.
- This problem is of special and crucial interest in understanding the geometry of orbital motion. The velocity of P is provided by taking the

cross product between it's position vector, $\boldsymbol{r}$, and the angular velocity vector, $\boldsymbol{\omega}$.

$$\boldsymbol{v} = \boldsymbol{\omega} \times \boldsymbol{r}$$

- Due to the planar nature of motion, a **polar** representation of variables will suffice. In that case, the position vector of P is $\boldsymbol{r} = r\boldsymbol{e}_r$ and the angular velocity vector is $\boldsymbol{\omega} = \omega\boldsymbol{e}_z$

- The velocity of P can then be determined by applying the cross product as

$$\begin{aligned} \boldsymbol{v} &= \omega\boldsymbol{e}_z \times r\boldsymbol{e}_r && (8) \\ &= \omega r(\boldsymbol{e}_z \times \boldsymbol{e}_r) && (9) \\ &= \omega r\boldsymbol{e}_\theta && (10) \end{aligned}$$

Derivatives of Polar Base Vectors

- By setting $\boldsymbol{v} = \dot{\boldsymbol{r}} = \frac{d(r\boldsymbol{e}_r)}{dt}$ into the above problem (where r remains constant) we obtain important relations about the time derivatives of the **polar** base vectors.

- For example, the time derivative of the **radial** base vector is found as

$$\dot{\boldsymbol{e}}_r = \frac{d(r\boldsymbol{e}_r)}{dt} = r\frac{d\boldsymbol{e}_r}{dt}$$

which yields

$$\dot{\boldsymbol{e}}_r = \omega\boldsymbol{e}_\theta.$$

- Similarly, an expression for the time derivative of the **circumferential** base vector can be obtained by using analogy with the above derivation

$$\begin{aligned} \dot{\boldsymbol{e}}_\theta &= \frac{d\boldsymbol{e}_\theta}{dt} && (11) \\ &= \boldsymbol{\omega} \times \boldsymbol{e}_\theta && (12) \\ &= \omega(\boldsymbol{e}_z \times \boldsymbol{e}_\theta) && (13) \\ &= -\omega\boldsymbol{e}_r && (14) \end{aligned}$$

Example: Orbital Motion of a Projectile

2.13 Consider a particle constrained to move in a circular orbit at a constant speed. Let $\mathbf{v}$ be the velocity at any instant. What is the acceleration of the particle; i.e., what is the vector $d\mathbf{v}/dt$?

Answer. The velocity vector $\mathbf{v}$ may be represented in polar coordinates as follows. Let $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\theta}}$, $\hat{\mathbf{z}}$, be, respectively, the unit vectors with origin at P in the directions of the radius, the tangent, and the polar axis perpendicular to the plane of the orbit. (See Fig. P2.13.) Then $\mathbf{v} = v\hat{\boldsymbol{\theta}}$, where v is the absolute value of $\mathbf{v}$. Hence, by differentiation,

$$\frac{d\mathbf{v}}{dt} = v\frac{d\hat{\boldsymbol{\theta}}}{dt} + \frac{dv}{dt}\hat{\boldsymbol{\theta}}.$$

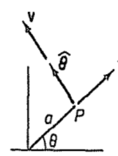


Figure P2.13 Velocity vector of a particle moving in a circular orbit.

(§2.3) The Summation Convention

Inquire GPT about the use and history of the topic.

> To whom is the summation convention attributed? What is it's relation with the contraction operation with some background about the latter?

- The summation convention is closely related with the **index contraction** operation in tensor algebra.

- For example, take the inner product between two vectors, $\boldsymbol{u}\cdot\boldsymbol{v}$. Written in it's matrix form this reads

$$\boldsymbol{u}\cdot\boldsymbol{v} = \{u_1\ u_2\ u_3\}\left\{\begin{array}{c} v_1 \\ v_2 \\ v_3 \end{array}\right\} = u_1v_1 + u_2v_2 + u_3v_3 = \sum_{i=1}^{3} u_iv_i$$

  which is essentially identical with the *scalar product* operation. The summation over a repeated pair of indices represents an *inner product* over those indices.

- Even when writing the vector as a sum over it's components and corresponding bases vectors, we use the index contraction process

$$\boldsymbol{u} = u_1\boldsymbol{e}_1 + u_2\boldsymbol{e}_2 + u_3\boldsymbol{e}_3 = \sum_{i=1}^{3} u_i\boldsymbol{e}_i$$

- The expression on the right hand of the second equal sign is said to be in *indicial notation.*
- Because contractions of indices occur very frequently in mechanics, Albert Einstein introduced the **summation convention**, whereby a summation sign is omitted and it is understood that whenever an index appears twice in a term, summation over that index is implied.
- Using the summation convention, the above equation simplifies to

$$\boldsymbol{u} = u_i \boldsymbol{e}_i$$

- In reading an expression written in indicial notation, the non-repeating indices are called as *free*, while the twice repeating ones are called as *dummy* indices.
- Note that, repetition of an index more than twice is **invalid** according to the summation convention.

Examples

Direction Cosines of a Vector

- Let $\boldsymbol{n}$ be a unit vector. It's components can be evaluated by using the single equation in indicial notation

$$n_i = \boldsymbol{n} \cdot \boldsymbol{e}_i = \cos(\boldsymbol{n}, \boldsymbol{e}_i)$$

where $(\boldsymbol{n}, \boldsymbol{e}_i) = \alpha_i$ are the angles that $\boldsymbol{n}$ makes with each coordinate axis.

- Using the fact that a unit vector has a magnitude equal to unity, we write

$$|\boldsymbol{n}|^2 = \boldsymbol{n} \cdot \boldsymbol{n} = n_i n_i = n_1 n_1 + n_2 n_2 + n_3 n_3 = 1$$

- Based on this expression a relation between the direction cosines can be obtained by expanding the dummy index

$$\cos\alpha_i \cos\alpha_i = \cos^2\alpha_1 + \cos^2\alpha_2 + \cos^2\alpha_3 = 1$$

Length of a fiber, $d\boldsymbol{x}$

- Consider a short fiber with components dx_1, dx_2, dx_3 in a three-dimensional Eucidean space with rectangular components x_1, x_2, x_3.
- The square of its length can be found by

$$ds^2 = dx_1^2 + dx_2^2 + dx_3^2$$

- In terms of its component representation the same result can be achieved by taking the scalar product of the vector by itself, as

$$\begin{aligned} ds^2 &= d\boldsymbol{x} \cdot d\boldsymbol{x} && (15) \\ &= (dx_i \boldsymbol{e}_i) \cdot (dx_j \boldsymbol{e}_j) && (16) \\ &= dx_i dx_j (\boldsymbol{e}_i \cdot \boldsymbol{e}_j) && (17) \\ &= dx_i \underbrace{dx_j \delta_{ij}}_{dx_i} && (18) \end{aligned}$$

where the definition of the Kroenecker's delta (§2.1) was used to convert the last line.

- In the above, the role of Kroenecker's delta is sometimes referred to be an **index shifter**.

The Cross Product in Indicial Notation

- Previously we saw the arisal of the alternator symbol in representing the vector product $\boldsymbol{u} \times \boldsymbol{v}$ as

$$\sum_k \sum_j \sum_i \epsilon_{ijk} u_i v_j \boldsymbol{e}_k$$

- But according to the summation convention there is no need to display the summation symbols and the cross product in indicial notation reads

$$\boldsymbol{u} \times \boldsymbol{v} = \epsilon_{ijk} u_i v_j \boldsymbol{e}_k$$

Higher (Second) Order Tensors

- The multiplication of a second order tensor with a vector also involves an inner product between the second (column) index of the second-order tensor and the index of the vector, reading

$$\boldsymbol{Au} = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \end{Bmatrix} = \begin{Bmatrix} A_{11}u_1 + A_{12}u_2 + A_{13}u_3 \\ A_{21}u_1 + A_{22}u_2 + A_{23}u_3 \\ A_{31}u_1 + A_{32}u_2 + A_{33}u_3 \end{Bmatrix} = A_{ij}u_j$$

- Note the placement of the *free* index (i), and the *repeated* indices (j). The repeated (*summation*) indices are also called as *dummy indices.*

About the Notation

- So far, we have seen two types of notations:
    1. Vector Notation, $\boldsymbol{f}, \boldsymbol{B}$, etc.
    2. Indicial Notation, f_i, B_{jk}, etc.
- There is another type of notation with which you (undergrads) are more familiar with, that is
    1. Matrix Notation, $\{f\}, [B]$, etc.
- All three types can be used to express (in other words 'to code') the axioms and equations of mechanics.

A Second Order Tensor in its Matrix Form

The rules of matrix algebra and the evaluation of determinants can be expressed more simply with the summation convention. An $m \times n$ *matrix* $\mathbf{A}$ is an ordered rectangular array of mn elements. We denote

$$\mathbf{A} = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \tag{2.3–9}$$

so that a_{ij} is the element in the ith row and jth column of the matrix $\mathbf{A}$. The index i takes the values $1, 2, \ldots, m$, and the index j takes the values $1, 2, \ldots, n$. A

The Transpose Operator

transpose of $\mathbf{A}$ is another matrix, denoted by $\mathbf{A}^T$, whose elements are the same as those of $\mathbf{A}$, except that the row numbers and column numbers are interchanged. Thus,

$$\mathbf{A}^T = (a_{ij})^T = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix} \tag{2.3–10}$$

The Matrix Product

The product of two 3 × 3 matrices $\mathbb{A} = (a_{ij})$, $\mathbb{B} = (b_{ij})$ is a 3 × 3 square matrix defined as

$$\mathbb{A}\cdot\mathbb{B} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix}$$
$$= \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & \cdots \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & \cdots \\ a_{31}b_{11} + a_{32}b_{21} + a_{33}b_{31} & \cdots \end{pmatrix} \tag{2.3–11}$$

whose element in the ith row and jth column can be written, with the summation convention, as

$$(\mathbb{A}\cdot\mathbb{B})_{ij} = (a_{ik}b_{kj}) \tag{2.3–12}$$

The Determinant

- The determinant of a second order tensor is the difference between the sum of triple products of components from each row/column having cyclic permutation with the sum having a reverse cyclic permutation:

$$\det \boldsymbol{A} = \det[A] = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \tag{19}$$
$$= \epsilon_{rst} A_{r1} A_{s2} A_{t3} \tag{20}$$
$$= \epsilon_{rst} A_{1r} A_{2s} A_{3t} \tag{21}$$

The $\epsilon - \delta$ Identity

The Kronecker delta and the permutation symbol are very important quantities that will appear again and again in this book. They are connected by the identity

$$\epsilon_{ijk}\epsilon_{ist} = \delta_{js}\delta_{kt} - \delta_{jt}\delta_{ks}. \qquad \blacktriangle \quad (2.3\text{–}19)$$

This ϵ-δ identity is used frequently enough to warrant special attention here. It can be verified by actual trial.

Problems

2.16 Show that

(a) $\delta_{ii} = 3$

(b) $\delta_{ij}\delta_{ij} = 3$

(c) $\epsilon_{ijk}\epsilon_{jki} = 6$

(d) $\epsilon_{ijk} A_j A_k = 0$

(e) $\delta_{ij}\delta_{jk} = \delta_{ik}$

(f) $\delta_{ij}\epsilon_{ijk} = 0$

2.17 Write Eqs. (2.1–1) and (2.1–5) in the index form, e.g., $\mathbf{u}\cdot\mathbf{v} = u_i v_i$.

Note. For Eq. (2.1–1), we may do the following: Define three unit vectors $\boldsymbol{\nu}^{(1)} = \mathbf{e}_1$, $\boldsymbol{\nu}^{(2)} = \mathbf{e}_2$, $\boldsymbol{\nu}^{(3)} = \mathbf{e}_3$; then $\mathbf{u} = u_i \cdot \boldsymbol{\nu}^{(i)}$.

2.18 Use the index form of vector equations to solve Probs. 2.5 through 2.9.

2.19 The vector product of two vectors $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$ is the vector $\mathbf{w} = \mathbf{u} \times \mathbf{v}$ whose components are

$$w_1 = u_2 v_3 - u_3 v_2, \qquad w_2 = u_3 v_1 - u_1 v_3, \qquad w_3 = u_1 v_2 - u_2 v_1.$$

Show that this can be shortened by writing

$$w_i = \epsilon_{ijk} u_j v_k.$$

2.20 Express Eqs. (2.1–7) in the index form.

Problem 2.22

- Derive the vector identity connecting three arbitrary vectors, $\boldsymbol{a}$, $\boldsymbol{b}$, and $\boldsymbol{c}$

$$\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) = (\boldsymbol{a} \cdot \boldsymbol{c})\boldsymbol{b} - (\boldsymbol{a} \cdot \boldsymbol{b})\boldsymbol{c}$$

by using the method of indicial notation.

(§2.4) Transformation Under Rotation

(§2.10) Partial Derivatives